PIC Compliance with the Technical Document and Blueprint

AlphaHelion Procurement Intelligence Cloud, 29 Oct 2026, code at commit `5a63ed3`. This is the summary; every requirement with its evidence is in PIC_Spec_Traceability_Matrix.md (490 rows).

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1. Answer in one paragraph

Not everything is complete, but the business processes are. Built and tested:

- every Blueprint workflow F1–F9 (84 steps; all 13 decision points tested both ways);
- the state machines, approval and tolerance tables (DMN), and segregation-of-duties rules;
- all purchase types and the five industry demo companies;
- 13 of 14 AI agents;
- the Axion signature case.

What remains:

- **Platform choices** where the Technical Document named Camunda, Kafka, ClickHouse, an LLM gateway and GraphQL, and the product uses simpler equivalents nobody has signed off yet.
- **Non-functional targets** that are not yet measured or not met: API latency, spend cube, MRP at 50,000 materials, 10,000 users, accessibility.
- **API and access conventions** applied only in part: idempotency keys, concurrency checks, attribute-based limits on commands.
- **Real external systems.** Today the ERP is a simulator and the other integrations wait for credentials.
- **Migration loaders** for your real ERP data.

2. Status counts

Document	Done	Partly done	Not done	Needs you (external)	Total
Technical Document	82 (46%)	68 (38%)	17 (10%)	10 (6%)	177
Blueprint (incl. 11 project design decisions)	199 (66%)	83 (28%)	12 (4%)	7 (2%)	301
Datasets D1–D11	9	2	0	1	12
All	**290**	**153**	**29**	**18**	**490**

The Blueprint describes the business, and two thirds of it is done. The Technical Document describes the platform and the quality targets, and that is where most of the "partly done" rows are.

3. What is complete

Area	Evidence
Workflows F1–F9	All 84 steps executed in the full data build; 13 of 13 gateways taken both ways; 389 backend tests
State machines and guards (TD §6)	Requisition, PO, receipt, inspection, SES, invoice, contract, sourcing, supplier, material request
DMN-01/02/03 and PO risk (TD §7)	Effective-dated, scoped rule sets plus an approval simulator
Purchase types (BP §22)	Stock, expense, service, capital asset, consignment, subcontracting, catalogue/punch-out
Industry packs (BP §23)	Construction, IT services, logistics, healthcare, general enterprise: one demo company each
Segregation of duties (BP §4.1)	SOD-01 to SOD-07 checked when the action is taken (agents excepted, §4)
Audit and multi-tenancy	Hash-chained audit verified per company; row-level security forced on 112 of 115 tables
Operations modules	Amendments, auctions, EDI, e-way bills, 4-way match, ERS, retention, freight audit, payment files, reconciliation, supplier master changes, invitations, punch-out, notifications, MFA, OIDC
Security hardening	Rate and body limits, security headers, session revocation, MFA lockout, privacy export/erasure, CI security scans
Code-vs-workflow review	89 findings fixed with tests (PIC_Code_Review_Report.md)

4. What is partly done or missing

4.1 Platform choices (decision needed: DEC-1)

Technical Document says	Built instead	Effect
NestJS microservices	FastAPI modular monolith	Simpler to run; services can be split later
Camunda 8 workflow engine	Own workflow engine in Postgres	All flows work; no BPMN modeller or viewer
Kafka/Avro event bus	Postgres outbox in the same process	Exactly-once and retries work; no external consumers
ClickHouse spend cube	Postgres	Fine at demo size; untested at 50 M lines
LiteLLM gateway	Rule-based Copilot and contract review	Answers follow the answer contract; no generative AI
GraphQL, a separate supplier-portal app	REST only; the portal is part of the same app	Same functions

4.2 Non-functional requirements (TD §15)

Requirement	Target	Now
API latency	P95 < 400 ms	Not measured (k6 scripts ready; needs staging)
Spend cube	< 3 s on 50 M lines	Not measured; the overview screen loads every line into memory
MRP	50,000 materials < 10 min	**681 s** best run (was about 8–9 h)
Scale	10,000 users, 5 M invoices a year	Not measured
Accessibility	WCAG 2.1 AA	Not audited
Audit retention	10 years	Default 8 years, configurable; no archive or purge job
Recovery	RPO 15 min / RTO 4 h	Runbook written; needs the Neon plan and a DR drill
Data residency	India by default	Neon Singapore (accepted for demo data)

4.3 Conventions and controls

- **Idempotency keys:** used on 26 of 138 POST routes, with no 24-hour expiry. Only purchase orders check `If-Match`.
- **Attribute-based access:** enforced on approvals only. The checks for company code, plant and value band exist but no command calls them yet.
- **Segregation of duties:** there is no mitigating-approval path, and the agent guard does not check SoD rules.
- **Scheduled jobs missing:** nightly reconciliation, payment-due reminders, monthly scorecards, retention and idempotency-record purges, scheduled MRP.

- **KPIs:** about 29 of 55 module KPIs are not yet computed, and no industry-template KPI is computed.
- **Front-end:**
  - **Missing screens and features:** savings pipeline (SPD-08), lifecycle strip, generic Approvals/Documents tabs, a BPMN process viewer, saved views, bulk actions.
  - **Bug:** the header always says "Manufacturing template".
- **Industry and purchase-type controls that exist only as flags or comments:**
  - Healthcare: clinical sign-off on awards.
  - Construction: advance recovery and bank guarantees.
  - Logistics: detention and demurrage.
  - Purchase types: ERS for scheduling agreements, Kanban/reorder-point planning, P-card.
  - Checks: the "last price + 5%" PO check, and risk constraints in award optimisation.
- **Organisation model:** no legal-entity or purchasing-org masters (hard-coded `1000` / `INR`). FX rates are stored but not yet used to convert amounts.
- **Tests:** no Playwright end-to-end tests, coverage gate, event contract tests or accessibility checks.

5. Datasets D1–D11

#	Dataset	Status
D1	Branch cases 9–34 (every gateway both ways)	Done
D1b	Operations module cases	Done
D2	Purchase types	Done
D3–D7	Construction, IT services, logistics, healthcare, general enterprise	Done
D8	Volume for NFRs	**Partly done.** Generator ready; only the 50,000-material MRP run was done (missed the target). 5 M invoices and 50 M spend lines never generated (needs staging)
D9	AI evaluation sets	Done. Two proposed gates fail: auto-apply precision 0.857 (target 0.95) and duplicate precision 0.83 (target 0.90). The targets need your sign-off
D10	Your real ERP masters and history	**Needs your ERP extract.** AXN history is generated; GMC/DBY came from the old prototype
D11	Reference data	**Partly done.** FX is one day only; no holiday calendar for the GMC/DBY plants; some 2027 state holidays not yet published

6. What you need to supply or decide

6.1 Decisions

#	Decision
DEC-1	Accept or reject the platform choices in §4.1
DEC-2	Hosting and data residency: Vercel (Pro or Hobby) plus Neon Singapore, or GCP Mumbai
DEC-3	Neon plan: point-in-time recovery ≥ 7 days, exports, DR region
DEC-4	Identity: your company login via OIDC, whether SAML is needed, switching MFA on (off in all companies today)
DEC-5	LLM gateway (URL, key, models), or accept the rule-based Copilot
DEC-6	Rule values: touchless band (default ₹25 L), approval band edges at exactly ₹2 L / ₹10 L, PO price-deviation rule, escalation (add manager or reassign), AI gate targets
DEC-7	Legal: retention per data category (10-year audit or 8), lawful-basis wording
DEC-8	Tax team: transport SAC 996791, implant GST rate, ERS self-billing under GST, job-work SAC 9988
DEC-9	Scope: multi-item sourcing events; which industry and purchase types the pilot uses first
DEC-10	Make the ZAP security scan blocking after triage

6.2 Systems and credentials

#	Supply
SUP-1	**SAP S/4HANA sandbox** (or your pilot ERP): the biggest blocker, because only a simulator exists today
SUP-2	GST IRP/GSP credentials
SUP-3	E-signature provider
SUP-4	Bank host-to-host / SFTP and the real paying account
SUP-5	EDI gateway and supplier EDI IDs; punch-out supplier accounts
SUP-6	Sanctions feed licence (and risk/news feeds if wanted)
SUP-7	Supplier Capability Digital Twin API
SUP-8	SMTP for e-mails
SUP-9	OCR engine for invoice capture

#	Supply
SUP-10	Hosting access (Vercel team/token or GCP project), GitHub secrets, a rotated Neon password
SUP-11	Tracing collector / log sink
SUP-12	A production-sized staging environment for the performance tests

6.3 Data and documents

#	Supply
DAT-1	**ERP extract**: 24–36 months of PO and invoice history, vendors with bank and tax IDs, materials, open contracts, open POs, open AP, GL totals
DAT-2	FX history and the currency of the legacy amounts
DAT-3	2027 holiday lists, holidays for the GMC/DBY plants, real IFSC codes, a fuel-index feed, your spend taxonomy or licensed UNSPSC
DAT-4	The Test-Case Workbook AHG/PRC/PIC/2026-10/03
DAT-5	FY 2027-28 budgets; NPPA ceiling prices for implants

6.4 Pending right now

- **Neon rebuild (option A):** prepared and rehearsed, but blocked by the permission check. Either run the command yourself, or approve it.
- **Neon password:** rotate it.

7. Work we can do without you

Size: S ≤ 1 day, M 2–5 days, L 1–3 weeks.

Rank	Work	Size
1	Idempotency keys on all POST routes (with 24 h expiry); `If-Match` on every editable document	M
2	Attribute-based limits on commands (company code, plant, value band); SoD mitigating approval; SoD inside the agent guard	M
3	MRP under 10 minutes at 50,000 materials; weekly planning buckets	M
4	Spend cube at scale: SQL aggregates, materialised views, filters and measures; run at 50 M lines	M–L
5	Scheduled jobs: reconciliation, payment due, scorecards, purges, scheduled MRP	S–M
6	Test layers: Playwright end-to-end, coverage gate, event contract tests, accessibility checks	M–L
7	The ~29 missing KPIs and the industry KPIs	M
8	Front-end gaps, and the "Manufacturing template" header fix (S)	M
9	Industry and purchase-type controls that are only flags today	M–L
10	Organisation masters (legal entity, purchasing org), configurable mandatory fields, FX conversion	M–L
11	Migration loaders for contracts, open POs and open AP, ready for your ERP extract	M
12	Remaining event consumers; a published event schema	S–M
13	Encryption key versioning and re-encryption; breached-password check; threat model	M
14	Data fixes: signature-case realised savings and first-pass match; holidays for the GMC/DBY plants	S

Together that is about 6–10 weeks for one engineer. Running in parallel, as in this phase, it is about 2–3 weeks.

8. How this was checked

- **Reviewer:** an independent read-only reviewer read both specs in full: Technical Document §1–§19 and Blueprint §2–§28 with Tables 24–28.
- **Each requirement** was checked against the code, the migrations, the tests and a fresh full data build (8 companies, every dataset pack). The project's own status documents were used only as pointers.
- **Spot checks:** four claims were re-checked independently before this report, and all four were confirmed:
    - idempotency on 26 of 138 POST routes;
    - the attribute checks are never called;
    - the header hard-codes "Manufacturing template";
    - MFA is off in all 8 companies.